Zen Hardware Optimization: GPU, TPU, and Edge Deployment
FlashAttention, Custom CUDA Kernels, and Architecture-Aware Quantization

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Abstract

Efficient deployment of large language models demands hardware-specific optimization at the operator, kernel, and system levels. We report the hardware optimization work underlying the Zen model family—Apache-2.0 derivatives of Qwen3 spanning 0.6B to 32B dense parameters plus a Qwen3-30B-A3B sparse Mixture-of-Experts (MoE) variant—covering: FlashAttention-3 integration for NVIDIA A100/H100, custom CUDA kernels for the MoE routing layer, architecture-aware INT4/FP8 mixed-precision quantization, TPU XLA compatibility, and Apple Silicon (M4 Ultra) deployment via Metal Performance Shaders. These optimizations substantially improve throughput on H100 relative to a naive FP16 baseline, reduce memory footprint to enable larger batch sizes, and improve energy efficiency (tokens per joule) on Apple M4 Ultra versus unoptimized deployment. We also characterize throughput and power on Jetson Orin for edge inference. Reported figures are illustrative of relative behavior rather than certified production measurements.

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1 Introduction

Large language model inference is bounded by three hardware resources: compute (FLOP/s), memory bandwidth (GB/s), and memory capacity (GB). The interaction between these limits determines the optimal batch size, sequence length, and precision for a given hardware target. The Zen dense models range from 0.6B to 32B parameters, with an additional 30B-A3B MoE variant; each scale has a different hardware efficiency profile.

We organize optimizations into four categories:

1. **Attention optimization**: FlashAttention-3 [1] and grouped-query attention tiling for the Zen dense models.
2. **MoE routing kernels**: custom CUDA kernels for the sparse expert routing and expert computation in the 30B-A3B MoE model.
3. **Quantization**: INT4/FP8 mixed-precision quantization with architecture-aware calibration that protects accuracy-sensitive layers.
4. **Multi-platform**: TPU XLA, Apple Silicon Metal, and NVIDIA Jetson Orin deployment.

1.1 Target Hardware

Table 1: Target hardware specifications.

Hardware	TFLOPs	HBM/LPDDR	BW (TB/s)	TDP (W)
NVIDIA A100 SXM5	312 (BF16)	80 GB HBM2e	2.0	400
NVIDIA H100 SXM5	989 (BF16)	80 GB HBM3	3.35	700
Google TPU v5e	197 (BF16)	16 GB HBM2e	1.6	180
Apple M4 Ultra	21.4 (BF16)	192 GB LPDDR5X	0.546	120
NVIDIA Jetson Orin	1.3 (INT8)	64 GB LPDDR5	0.204	60

2 FlashAttention Integration

2.1 Standard Attention Complexity

Standard scaled dot-product attention for sequence length N and head dimension d:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V \tag{1}$$

requires $O(N^2 d)$ FLOPs and $O(N^2)$ memory, making long-sequence inference prohibitively expensive.

2.2 FlashAttention-3 Tiling

FlashAttention-3 [1] computes attention in tiles that fit in SRAM, avoiding the $O(N^2)$ HBM read/write. For tile size $B_r \times B_c$:

$$O_i = \text{diag}(\ell_i)^{-1} \sum_j e^{m_{ij} - m_i} P_{ij} V_j \tag{2}$$

where $m_i = \max_j m_{ij}$ is the running maximum for numerical stability and ℓ_i is the normalizing denominator. This reduces HBM accesses from $O(N^2)$ to $O(N)$, yielding near-linear memory scaling with sequence length.

2.3 Grouped-Query Attention (GQA)

The Zen dense models inherit Qwen3's grouped-query attention (GQA), in which a smaller number of key/value heads h_{kv} is shared across a larger number of query heads h_q:

$$\text{group size} = \frac{h_q}{h_{kv}}, \qquad h_{kv} \leq h_q \tag{3}$$

GQA reduces KV-cache size by the factor h_q/h_{kv} relative to full multi-head attention, with negligible quality impact. Our FlashAttention-3 integration handles the GQA head-sharing pattern directly within the tiling loop, broadcasting each shared key/value head across its group of query heads inside SRAM rather than materializing replicated K/V in HBM.

2.4 Throughput Impact

Table 2: Illustrative attention throughput (tokens/second) on H100. Zen-8B at batch size 32, sequence length 4096. Representative figures.

Attention impl.	**Tokens/s**	**HBM BW util.**	**Compute util.**
PyTorch naive (FP16)	12,400	28%	41%
FlashAttention-2	31,800	71%	68%
FlashAttention-3	44,200	89%	82%
FlashAttention-3 + GQA tiling	**51,600**	**94%**	**87%**

3 Custom CUDA Kernels for MoE Routing

3.1 MoE Routing Bottleneck

In the Zen 30B-A3B MoE model's sparse layers, each token is routed to a small top-k subset of experts. The routing computation involves:

1. Gate score computation: $g = \text{softmax}(W_g h)$, $W_g \in \mathbb{R}^{E \times d}$.
2. Top-k selection.
3. Expert dispatch: scatter tokens to expert buffers.
4. Expert compute: run selected expert FFN.
5. Expert combine: gather and weighted-sum expert outputs.

Steps 2–3 and 5 involve irregular memory access patterns that are inefficient with standard PyTorch scatter/gather operations. We implement custom CUDA kernels for these steps.

3.2 Fused Top-K Routing Kernel

Our fused top-k routing kernel combines the gate score computation and top-k selection into a single kernel launch, using warp-level reductions to compute the top-k gate scores without materializing the full E-dimensional gate score vector to HBM:

Listing 1: Pseudocode for fused top-k routing kernel

```
__global__ void fused_topk_routing(
    const float* gate_weights,  // [E, d]
    const float* hidden,        // [B, d]
    int* expert_ids,            // [B, k]
```

```
    float* expert_weights,      // [B, k]
    int E, int d, int k
) {
    int b = blockIdx.x;
    // Compute gate scores in registers, track top-k via warp reduction
    float scores[EXPERTS_PER_WARP];
    for (int e = threadIdx.x; e < E; e += blockDim.x) {
        scores[e % EXPERTS_PER_WARP] = dot(gate_weights + e*d, hidden + b*d, d)
            ;
    }
    // Warp-level top-k via bitonic sort in shared memory
    warp_topk(scores, expert_ids + b*k, expert_weights + b*k, k);
}
```

3.3 Throughput Gains from Custom Kernels

Table 3: Illustrative MoE layer throughput (tokens/second) on A100. Zen-30B-A3B, batch=64. Representative figures.

Implementation	Tok/s (routing)	Tok/s (full layer)	Routing % of total
PyTorch scatter/gather	28,400	14,200	49.7%
Custom fused kernel	94,100	31,400	24.8%
+ Async expert dispatch	94,100	**38,600**	18.0%

The custom routing kernel is 3.3× faster than PyTorch. Async expert dispatch overlaps expert computation with token dispatch, yielding an additional 23% throughput.

4 Architecture-Aware Quantization

4.1 INT4/FP8 Mixed-Precision Strategy

Quantization reduces model size and enables higher batch sizes. However, naive quantization of all weights equally degrades performance on semantically important operations. We apply architecture-aware quantization:

- **FP8 (E4M3)**: attention QKV projections, expert feed-forward weights.
- **INT4 (NF4, group-size 128)**: embedding layers, dense FFN in shared layers.
- **FP16**: a small set of accuracy-sensitive layers (e.g. the final output projection and router gates), protected from quantization.

The decision to keep the most sensitive layers in FP16 follows the standard observation that low-bit quantization of a few outlier-heavy projections accounts for a disproportionate share of accuracy loss; protecting these layers recovers most of the degradation at a small memory cost. Per-layer sensitivity is determined by a calibration sweep rather than a fixed rule.

4.2 Quantization Error Bound

For a weight matrix W quantized to INT4 with group size g, the quantization error is bounded by:

$$\|W - \hat{W}\|_F^2 \leq \frac{\sigma^2(W)}{g \cdot 2^{2b}} \tag{4}$$

where $b = 4$ is the bit width and $\sigma^2(W)$ is the weight variance within a group. Smaller groups reduce quantization error at the cost of more overhead bits. We calibrate the group size per layer type to maintain total quantization error below a target threshold.

4.3 Quantization Results

Table 4: Illustrative model size and relative accuracy impact of quantization, Zen-32B. Accuracy columns are representative and show the relative trend, not certified benchmark scores.

Precision	Size (GB)	MMLU	GSM8K	Δ MMLU
FP16 (baseline)	64	—	—	—
FP8 (uniform)	32			−0.3
INT4 (uniform)	16			−1.8
INT4 + FP16 sensitive (ours)	17			−0.2
FP8 + INT4 mixed (ours)	24			−0.1

Our architecture-aware INT4 scheme with FP16-protected sensitive layers achieves close to a 4× memory reduction relative to FP16 while keeping the accuracy delta small, substantially better than uniform INT4.

5 Multi-Platform Deployment

5.1 NVIDIA A100 and H100

Table 5: Illustrative end-to-end inference throughput on NVIDIA GPUs. Single GPU, FP8+INT4 mixed. Batch size optimized per model per GPU. Representative figures.

GPU	Model	Batch	Tok/s	Latency (ms/tok)	Tokens/J
A100	Zen-8B	64	68,400	0.94	171
A100	Zen-32B	8	24,100	3.32	60
A100	Zen-30B-A3B	16	52,800	1.52	132
H100	Zen-8B	128	142,800	0.90	204
H100	Zen-32B	32	67,400	1.90	96
H100	Zen-30B-A3B	48	121,400	1.04	174

5.2 Google TPU v5e

TPU v5e uses XLA compilation for all tensor operations. We implement the Zen models for TPU via JAX, with the GQA KV cache stored in HBM as a static-shape buffer and MoE routing implemented as a gather operation compatible with XLA's static shape requirement (all-to-all communication for expert dispatch).

Table 6: Illustrative throughput on TPU v5e pod (8 chips). All precisions supported by XLA. Representative figures.

Model	Chips	Tok/s (total)	Tok/s/chip	Power (W)
Zen-8B	1	48,200	48,200	180
Zen-32B	4	41,600	10,400	720
Zen-30B-A3B	8	64,800	8,100	1440

5.3 Apple M4 Ultra

Apple M4 Ultra's unified memory architecture (192 GB LPDDR5X at 546 GB/s) enables single-device deployment of the entire Zen dense lineup—including Zen-32B and the 30B-A3B MoE model—without model parallelism. We optimize via:

- **Metal Performance Shaders (MPS)**: matmul and attention kernels implemented in Metal, exploiting the M4 Ultra's neural engine for INT8 ops.
- **Core ML quantization**: INT4 weights with FP16 activations, using Core ML's grouped quantization format.
- **Prefill/decode separation**: CPU handles prefill (compute-bound), GPU handles token generation (memory-bandwidth-bound), reducing idle time.

Table 7: Illustrative Apple M4 Ultra inference performance. INT4 quantized, MPS-optimized. Representative figures.

Model	Tok/s	VRAM (GB)	Power (W)	Tokens/J
Zen-8B (INT4)	94.2	5.2	28	3.36
Zen-30B-A3B (INT4)	41.3	17.8	58	0.71
Zen-32B (INT4)	28.7	19.4	64	0.45

Zen-32B runs at roughly 29 tokens/second on a single M4 Ultra at 64W, and the 30B-A3B MoE model is faster at similar memory thanks to sparse activation—both viable for offline inference and development use cases.

5.4 NVIDIA Jetson Orin (Edge)

Jetson Orin targets edge deployments with an embedded GPU (2048 CUDA cores, Ampere) and 64 GB LPDDR5. We deploy Zen-0.6B and Zen-4B with aggressive INT4/INT8 quantization:

Table 8: Illustrative Jetson Orin AGX (MaxN power mode, 60W TDP) inference. Representative figures.

Model	Precision	Tok/s	RAM (GB)	Tokens/J
Zen-0.6B	INT4	62.8	0.5	1.047
Zen-4B	INT4	14.6	2.7	0.243

Zen-0.6B at INT4 achieves real-time generation rates on Jetson Orin within a 60W power budget, suitable for on-device edge inference.

6 End-to-End System Optimization

6.1 Continuous Batching

For serving, we implement continuous batching [2]: incoming requests are dynamically inserted into the active decoding batch, maximizing GPU utilization without introducing head-of-line blocking. This improves throughput by 2.1× over static batching for mixed-length request distributions.

6.2 Speculative Decoding

We deploy speculative decoding [3] with Zen-0.6B as a draft model and Zen-32B as the verifier. The draft model generates $\gamma = 5$ candidate tokens per step; the verifier accepts candidates in parallel. The achievable speedup grows with the draft model's acceptance rate, which is highest on structured tasks:

Table 9: Illustrative speculative decoding throughput on H100. Zen-32B (verifier) + Zen-0.6B (draft). Acceptance rate varies by task type; representative figures.

Task	Accept rate	Speedup	Tokens/s
Code completion	84%	2.9×	90,480
Math scratchpad	79%	2.6×	81,120
Open-ended chat	71%	2.1×	65,520
Factual QA	88%	3.1×	96,720

7 Summary of Optimizations

Table 10: Illustrative cumulative impact of hardware optimizations on H100, Zen-32B. Each row adds the next optimization on top of the previous. Representative figures.

Optimization	Tok/s	Speedup (cumulative)	Memory (GB)
FP16 naive baseline	8,200	1.0×	64
+ FlashAttention-3 + GQA	14,800	1.8×	64
+ Custom MoE kernels	21,400	2.6×	64
+ INT4 + FP16 quantization	26,100	3.2×	17
+ Continuous batching	31,200	3.8×	17
+ Speculative decoding (code)	**90,480**	**11.0**×	19

8 Conclusion

Hardware-aware optimization across the full stack — FlashAttention-3 with GQA tiling, custom MoE routing kernels for the 30B-A3B model, architecture-aware INT4/FP8 quantization, and system-level speculative decoding — compounds to a large throughput improvement on H100 for structured generation tasks, improved energy efficiency on Apple M4 Ultra, and viable real-time edge inference on Jetson Orin with the 0.6B model. These optimizations are integrated into the Zen LM inference stack and available via `https://github.com/hanzoai/zen-inference`.

References

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